

# **Proofs and Problem Solving**

## **Week 10: Permutations**

**Notes based on Martin Liebeck's**

***A Concise Introduction to Pure Mathematics***



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# Week 10: Permutations

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## 16. Permutations

Two weeks ago, we looked at functions  $f : X \rightarrow Y$  between sets  $X$  and  $Y$ . This week, we're going to focus on bijective functions whose domain  $X$  and codomain  $Y$  are both the same finite set. In other words, we're going to study bijections from a finite set to itself. Functions of this kind are called "permutations".

Since it makes no difference what the elements of a set happen to be called, we will assume that the elements of our set are called  $1, 2, \dots, n$ . So, we're looking at bijections  $\{1, 2, \dots, n\} \rightarrow \{1, 2, \dots, n\}$ .

### 16.1 Permutations

**Definition 16.1** Given  $n \in \mathbb{N}$ , denote by  $S_n$  the set of all bijections

$$\{1, 2, \dots, n\} \longrightarrow \{1, 2, \dots, n\}.$$

We call elements of  $S_n$  **permutations** of the set  $\{1, 2, \dots, n\}$ .

As we saw last week, permutations are important for combinatorics and counting. They are also interesting from an algebraic perspective, as we'll see below.

**Example 16.1** The set  $S_1$  consists of one bijection  $f : S_1 \rightarrow S_1$  defined by  $f(1) = 1$ .

The set  $S_2$  consists of two bijections:

- a function  $f$  such that  $f(1) = 1$  and  $f(2) = 2$  (the identity function),
- a function  $g$  such that  $g(1) = 2$  and  $g(2) = 1$ .

We know exactly how many permutations of  $\{1, 2, \dots, n\}$  there are.

**Lemma 16.1** The set  $S_n$  consists of exactly  $n!$  permutations.

*Proof.* Permutations in  $S_n$  are just ways of rearranging the numbers  $\{1, 2, \dots, n\}$  written in order. That is, we can identify each rearrangement of numbers  $\{1, 2, \dots, n\}$  written in order

with the unique function which maps  $\{1, 2, \dots, n\}$  onto that rearrangement. ■

Notice that an element of the set  $S_n$  is a *function*. So, we are dealing with the whole function,  $f$ , and not just the value of the function  $f(k)$  for some number  $k$ .

## 16.2 Table notation

Here we will introduce some useful notation for denoting permutations. To represent a permutation  $f \in S_n$ , we just need to remember where each integer  $k \in \{1, 2, \dots, n\}$  gets sent to under  $f$  and then we can then write this out using a matrix:

$$\begin{pmatrix} 1 & 2 & 3 & \cdots & n \\ f(1) & f(2) & f(3) & \cdots & f(n) \end{pmatrix}$$

where the first row represents the domain of  $f$  and the bottom row represents where each value is sent to under  $f$ . For example,

$$\begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$$

is the permutation in  $S_3$  defined by  $f(1) = 3$ ,  $f(2) = 1$ , and  $f(3) = 2$ .

### ■ Exercise 16.1

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 \\ 7 & 5 & 1 & \star & 9 & 2 & 10 & 4 & 3 & 8 \end{pmatrix}$$

denotes a permutation in  $S_{10}$ . What is  $\star$ ?

## 16.3 Composition and identity

Take two permutations  $f \in S_n$  and  $g \in S_n$ . Since the domain of  $f$  matches the codomain of  $g$ , we can form the composite function which (by definition of composition) is given by  $(f \circ g)(k) = f(g(k))$  for each  $k \in \{1, 2, \dots, n\}$ .

$$f \circ g : \{1, 2, \dots, n\} \rightarrow \{1, 2, \dots, n\},$$

$$\{1, 2, \dots, n\} \xrightarrow{g} \{1, 2, \dots, n\} \xrightarrow{f} \{1, 2, \dots, n\}.$$

The composite of two bijections is again a bijection, so  $f \circ g$  is a permutation too.

We will usually drop the  $\circ$  and just write  $fg$  instead of  $f \circ g$ .

**Example 16.2** Let

$$f = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 1 & 4 \end{pmatrix}, \quad \text{and} \quad g = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix}$$

We can determine  $fg$  by plugging in each number  $k \in \{1, 2, 3, 4\}$  and seeing what  $g(k)$  and then  $f(g(k))$  are:

$$\left( \begin{array}{c|cccc} k & 1 & 2 & 3 & 4 \\ g(k) & 4 & 3 & 2 & 1 \\ f(g(k)) & 4 & 1 & 3 & 2 \end{array} \right)$$



Thus, taking the top and bottom rows, we see that

$$fg = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 3 & 2 \end{pmatrix}$$

We use the symbol  $\iota$  (the Greek letter iota) to denote the *identity permutation*

$$\iota = \begin{pmatrix} 1 & 2 & 3 & 4 & \dots & n \\ 1 & 2 & 3 & 4 & \dots & n \end{pmatrix}, \quad \iota(x) = x \quad \text{for all } x \in \{1, 2, \dots, n\}.$$

For every  $f \in S_n$ , we have  $\iota \circ f = f \circ \iota = f$ , that is,  $\iota f = f \iota = f$ .

Another reason we drop the  $\circ$  from the above notation is so that we can think of composing functions as if we are multiplying elements together, and the function  $\iota$  is playing the role of the number 1: when you multiply  $\iota$  by any other element, you get the same element back. However, unlike multiplication for integers, composition of functions is not commutative.

**Example 16.3** Let us consider composition of two permutations in  $S_5$ :

$$f = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 4 & 5 & 1 \end{pmatrix} \quad \text{and} \quad g = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 1 & 3 & 4 & 5 \end{pmatrix}.$$

Then

$$fg = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 2 & 4 & 5 & 1 \end{pmatrix} \quad \text{but} \quad gf = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 3 & 4 & 5 & 2 \end{pmatrix}.$$

That is,  $fg \neq gf$ .

*Sometimes* it happens that  $fg = gf$ ; for example, can you show that  $fg$  is always equal to  $gf$  when  $f, g \in S_2$ ? But when  $n \geq 3$ , there are always some  $f, g \in S_n$  such that  $fg \neq gf$ .

## 16.4 Inverse permutations

Let  $f$  be any permutation in  $S_n$ . Then

$$1 \mapsto f(1), 2 \mapsto f(2), 3 \mapsto f(3), \dots, n \mapsto f(n),$$

and, as  $f$  is injective, all the numbers  $f(1), f(2), f(3), \dots, f(n)$  are different.

Since  $f \in S_n$  is bijective, the inverse permutation  $f^{-1}$  always exists with

$$ff^{-1} = \iota = f^{-1}f.$$

To figure out what  $f^{-1}$  is given  $f$ , note that the permutation  $f^{-1}$  maps

$$f(1) \mapsto 1, f(2) \mapsto 2, f(3) \mapsto 3, \dots, f(n) \mapsto n.$$

Thus, if we have our matrix representing  $f$ ,

$$f = \begin{pmatrix} 1 & 2 & 3 & \dots & n \\ f(1) & f(2) & f(3) & \dots & f(n) \end{pmatrix},$$

then the inverse is

$$f^{-1} = \begin{pmatrix} f(1) & f(2) & f(3) & \dots & f(n) \\ 1 & 2 & 3 & \dots & n \end{pmatrix}.$$

Since  $f(1), f(2), \dots, f(n)$  runs through each integer  $1, 2, \dots, n$  exactly once, the top row is just a rearrangement of  $\{1, 2, \dots, n\}$ , so if we rearrange the columns so that the top row is in order, then we will get

$$f^{-1} = \begin{pmatrix} 1 & 2 & 3 & \dots & n \\ f^{-1}(1) & f^{-1}(2) & f^{-1}(3) & \dots & f^{-1}(n) \end{pmatrix}.$$

**Example 16.4** Let  $f = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 4 & 5 & 1 \end{pmatrix} \in S_5$ . Find  $f^{-1}$ .

First flip the matrix (switching the bottom and top rows) and then reorder the columns so that the top is ordered from 1 to 5:

$$f^{-1} = \begin{pmatrix} 2 & 3 & 4 & 5 & 1 \\ 1 & 2 & 3 & 4 & 5 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 1 & 2 & 3 & 4 \end{pmatrix}.$$

Let us summarize some of the properties about permutations in  $S_n$  we have proved.

**Lemma 16.2** The set  $S_n$  equipped with the composition rule  $\circ$  has the following properties:

- **Closure:** for any  $f$  and  $g$  in  $S_n$ , one has  $f \circ g \in S_n$ ,
- **Associativity:** for any  $f, g$ , and  $h$  in  $S_n$ , one has

$$f \circ (g \circ h) = (f \circ g) \circ h,$$

- **Identity element:** there is unique permutation  $\iota \in S_n$  such that

$$f \circ \iota = \iota \circ f = f$$

for any  $f \in S_n$  (and we call  $\iota$  the identity permutation),

- **Inverse:** for any  $f \in S_n$ , there is unique  $f^{-1} \in S_n$  such that

$$f \circ f^{-1} = f^{-1} \circ f = \iota.$$

In *Fundamentals of Pure Math* (FPM), you will learn more about *groups*, which are sets equipped with the above structure, that is, a way of composing elements that satisfies the above properties. We have seen several examples so far:

- real numbers equipped with  $+$ ,
- non-zero real numbers equipped with  $\cdot$  (i.e. multiplication),
- complex numbers equipped with  $+$ ,
- non-zero complex numbers equipped with  $\cdot$ ,
- $n$ th roots of unity equipped with  $\cdot$ ,
- $\mathbb{Z}_n$  equipped with  $+$ ,
- $\mathbb{Z}_n \setminus \{0\}$  equipped with  $\cdot$  if  $n$  is prime (in which case every element has an inverse).

We can now add  $S_n$  equipped with composition (also known as the *permutation group*) to this list.

## 16.5 Cycles

Beyond using a matrix representation, there is also a second very common way to write a permutation.

**Definition 16.2** Let  $r \geq 1$  and let  $a_1, a_2, \dots, a_r$  be distinct elements of the set  $\{1, 2, \dots, n\}$ . Let  $f \in S_n$  be the permutation such that

$$f(a_1) = a_2, f(a_2) = a_3, \dots, f(a_{r-1}) = a_r, f(a_r) = a_1,$$

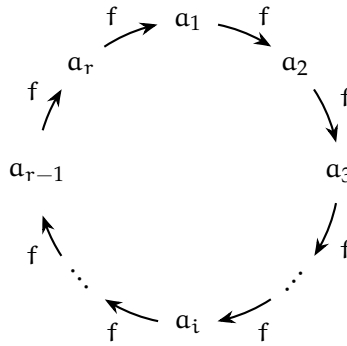
and, for  $k \notin \{a_1, a_2, \dots, a_r\}$ ,

$$f(k) = k.$$

Then we call  $f$  a *cycle* or *cyclic permutation* (of length  $r$ ) and denote it by

$$f = (a_1 \ a_2 \ \dots \ a_r).$$

We call these "cycles" since, if you imagine arranging the numbers  $a_1, \dots, a_r$  around a circle,  $f$  just rotates these numbers. In other words, this is the permutation that sends  $a_1$  to  $a_2$ ,  $a_2$  to  $a_3$ , ..., and  $a_r$  to  $a_1$ , and leaves everything else alone.



The points that are left alone are called a *fixed point* of the permutation  $f \in S_n$ : i.e. a fixed point is an element  $a \in \{1, \dots, n\}$  such that  $f(a) = a$ .

Notice that if  $j \in \{1, 2, \dots, n\}$ , then  $(j) = \iota$ , so  $\iota$  is a cycle of length 1. Length 1 cycles are often called *trivial cycles*.

**Example 16.5** To find the matrix representation of the cycle  $f = (2 \ 5 \ 4 \ 7)$ , observe that for  $k \notin \{2, 5, 4, 7\}$ ,  $f(k) = k$ . Moreover, the cycle notation tells us that  $f(2) = 5$ ,  $f(5) = 4$ ,  $f(4) = 7$  and  $f(7) = 2$ . Thus,  $f$  has matrix representation  $f = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 1 & 5 & 3 & 7 & 4 & 6 & 2 \end{pmatrix} \in S_7$ .

Note that there is some ambiguity with cycle notation and which set  $S_n$  it belongs to: we do not automatically know what  $n$  is. In the above example, the cycle notation only told us  $n \geq 7$ . When we wrote the permutation in matrix representation, we fixed  $n = 7$ . Taking  $n$  to be minimal is a standard assumption but is still an assumption!

**Example 16.6** Let  $f = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 5 & 3 & 2 & 4 & 6 \end{pmatrix} \in S_6$ . If we are given that  $f$  is a cycle, then we can figure out its cycle notation by first finding the values  $a_1, \dots, a_r$  that  $f$  cycles through, that is, the values  $k$  for which  $f(k) \neq k$ . These are 2, 5, and 4, and  $f$  leaves all other elements of  $\{1, 2, 3, 4, 5, 6\}$  fixed (that is  $f(k) = k$  for  $k \notin \{2, 5, 4\}$ ). Then we just look at where  $f$  sends these numbers in succession:

$$f(2) = 5, f(5) = 4, f(4) = 2,$$

Thus,  $f = (2\ 5\ 4)$ .

Notice that the notation of a cycle is not unique. Indeed, for a cycle  $(a_1\ a_2\ \cdots\ a_r) \in S_n$ , we have

$$(a_1\ a_2\ \cdots\ a_r) = (a_2\ \cdots\ a_r\ a_1) = (a_3\ \cdots\ a_r\ a_1\ a_2) = \cdots. \quad (16.5.1)$$

**Example 16.7** Let  $f = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 1 & 5 & 3 & 7 & 4 & 6 & 8 & 2 \end{pmatrix} \in S_9$ . Then

$$f = (2\ 5\ 4\ 7\ 8) = (5\ 4\ 7\ 8\ 2) = (4\ 7\ 8\ 2\ 5) = (7\ 8\ 2\ 5\ 4) = (8\ 2\ 5\ 4\ 7).$$

Recall that  $S_n$  is not commutative for  $n \geq 3$ , that is, we can find  $f, g \in S_n$  so that  $fg \neq gf$ . However, if  $f$  and  $g$  are cycles that cycle through disjoint sets of numbers, then we do have  $fg = gf$ . We make this more precise below:

**Definition 16.3** If  $f = (a_1\ a_2\ \cdots\ a_r)$  and  $g = (b_1\ b_2\ \cdots\ b_s)$  where  $\{a_1, \dots, a_r\}$  and  $\{b_1, \dots, b_s\}$  are disjoint sets, we say that  $f$  and  $g$  are *disjoint* in  $S_n$ .

**Proposition 16.3** If  $f$  and  $g$  are disjoint cycles in  $S_n$ , then  $fg = gf$ .

*Proof.* Let  $f = (a_1\ a_2\ \cdots\ a_r)$  and  $g = (b_1\ b_2\ \cdots\ b_s)$  where  $\{a_1, \dots, a_r\}$  and  $\{b_1, \dots, b_s\}$  are disjoint sets in  $\{1, 2, \dots, n\}$ . Let us look at the charts for  $f$  and  $f \circ g$ , but ordering the first row so that they begin with  $a_1, \dots, a_r, b_1, \dots, b_s$ :

$$\left( \begin{array}{c|cccccccccc} k & a_1 & a_2 & \cdots & a_{r-1} & a_r & b_1 & b_2 & \cdots & b_{s-1} & b_s & \cdots \\ g(k) & a_1 & a_2 & \cdots & a_{r-1} & a_r & b_2 & b_3 & \cdots & b_s & b_1 & \cdots \\ f(g(k)) & a_2 & a_3 & \cdots & a_r & a_1 & b_2 & b_3 & \cdots & b_s & b_1 & \cdots \end{array} \right)$$

Above, the final numbers in " $\cdots$ " are the remaining integers  $\{1, 2, \dots, n\}$  not in either  $\{a_1, \dots, a_r\}$  or  $\{b_1, \dots, b_s\}$ , and for  $k$  in this range, we have  $f(k) = g(k) = k$ . Notice that since the  $a$ 's are not in  $\{b_1, \dots, b_s\}$ ,  $g$  fixes the numbers  $a_1, \dots, a_r$  (that is,  $g(a_i) = a_i$  for  $i = 1, \dots, r$ ) and then cycles through the  $b$ 's. Similarly, since the  $b$ 's are not in  $\{a_1, \dots, a_r\}$ ,  $f$  fixes the numbers  $b_1, \dots, b_s$  (that is,  $f(b_i) = b_i$  for  $i = 1, \dots, s$ ) and then cycles through the  $a$ 's. Thus, we get that

$$fg = \begin{pmatrix} a_1 & a_2 & \cdots & a_{r-1} & a_r & b_1 & b_2 & \cdots & b_{r-1} & b_r & \cdots \\ a_2 & a_3 & \cdots & a_r & a_1 & b_2 & b_3 & \cdots & b_r & b_1 & \cdots \end{pmatrix}.$$

Notice that if we did the exact same computation but with  $gf$  instead of  $fg$ , we would get the same matrix representation, thus  $fg = gf$ . ■

## 16.6 Order of a Permutation

Let  $f$  be any permutation in  $S_n$  and let  $m$  be any positive integer. Then we define

$$f^m = \underbrace{f \circ f \circ f \circ \cdots \circ f \circ f}_{m \text{ times}}.$$

We also define  $f^{-m} = (f^{-1})^m$  and  $f^0 = \iota$ . Then we have the familiar power rules for compositions as we do for multiplication: for  $r, s \in \mathbb{Z}$ ,

$$f^r f^s = f^{r+s} \text{ and } (f^r)^s = f^{rs}.$$

**Example 16.8** Let  $f = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 4 & 5 & 1 \end{pmatrix} \in S_5$ . Then, one can check

$$f^{-1} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 1 & 2 & 3 & 4 \end{pmatrix}, f^0 = \iota, f^1 = f, f^2 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 1 & 2 \end{pmatrix},$$

$$f^3 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 5 & 1 & 2 & 3 \end{pmatrix}, f^4 = f^{-1}, f^5 = \iota, f^6 = f, f^7 = f^2, \dots$$

■ **Exercise 16.2** Let  $f = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 1 & 3 & 4 & 5 \end{pmatrix} \in S_5$ . What is  $f^{2019}$ ?

The set  $S_n$  of permutations of  $\{1, \dots, n\}$  has only finitely many elements ( $n!$  of them, as we have seen). So if we take a permutation  $f \in S_n$  and look at the sequence  $f, f^2, f^3, \dots$ , eventually this must repeat itself (otherwise we would have infinitely many distinct permutations of  $n$  integers). In other words, if we have a deck of  $n$  cards, and we shuffle the cards according to some permutation  $f$ , then no matter how random the shuffle is, so long as we use the *same* shuffle  $f$ , repeating it over and over will eventually return the cards to their original order.

**Definition 16.4** Let  $f$  be a permutation in  $S_n$ . The smallest positive integer  $m$  such that

$$f^m = \iota$$

is called the *order* of the permutation  $f$ .

Note that if  $f$  is a cycle of length  $r$  in  $S_n$ , then the powers loop around themselves and follow the following pattern:

$$\dots f^{-1} = f^{r-1}, \underbrace{\iota = f^0, f^1, f^2, f^3, \dots, f^{r-1}}, f^r = \iota, f^{r+1} = f, \dots$$

where the permutations  $f^0, f^1, \dots, f^{r-1}$  are all distinct. To prove this, consider the following three cases for the cycle  $f = (a_1 \cdots a_r)$ :

- If  $0 \leq k < r$ , then  $f^k(a_i) = a_{i+k}$ , which ranges through  $a_1, a_2, \dots, a_r$  as  $k = 0, 1, \dots, r-1$  and these are all distinct numbers.
- Note  $f^r(a_1) = f \circ f^{r-1}(a_1) = f(a_r) = a_1$ , where we have looped around the whole cycle. Moreover, as it does not matter which element starts our cycle (as in (16.5.1)), then  $f^r(a_i) = a_i$  for all  $i = 1, \dots, r$ . Since  $f(b) = b$  for all  $b \notin \{a_1, \dots, a_r\}$ , this means  $f^r(b) = b$ , and thus  $f^r = \iota$ .
- If  $k > r$  or  $k < 0$ , then, by the remainder theorem, we can write  $k = qr + m$  where  $0 \leq m < r$  is a remainder and  $q \in \mathbb{Z}$  is some quotient. Then we have

$$f^k = f^{qr+m} = (f^r)^q f^m = \iota^q f^m = f^m.$$

By the first bullet point, this is one of the distinct elements  $f^0, f^1, \dots, f^{r-1}$ . We have thus shown the following.

**Lemma 16.4** If  $a_1, \dots, a_r$  are distinct numbers in  $\{1, 2, \dots, n\}$  and  $f = (a_1 a_2 \cdots a_r)$  is a permutation, then the order of  $f$  is  $r$ .

More generally, for any permutation  $f \in S_n$ , we can show the following:

**Lemma 16.5** Let  $f \in S_n$  have order  $m$ . If  $f^k = \iota$  for some integer  $k \in \mathbb{N}$ , then  $m$  divides  $k$ .

*Proof.* If  $m$  is the order of  $f \in S_n$  and  $f^k = \iota$ , then  $m \leq k$  by definition of  $m$ . By the remainder theorem, there are  $q \geq 0$  and  $0 \leq r < m$  such that  $k = qm + r$ , and so

$$f^k = f^{qm+r} = (f^m)^q f^r = (\iota)^q f^r = \iota f^r = f^r.$$

Recall that  $m$  is the smallest positive integer for which  $f^m = \iota$ , but now we have  $r \in \{0, 1, \dots, m-1\}$  (and so  $r < m$ ) for which  $f^r = \iota$ . This is only possible if  $r$  is not positive, i.e. if  $r = 0$ . Thus,  $k = qm + 0 = qm$ , giving  $m|k$ . ■

## 16.7 Cycle notation

Not every permutation is a cycle. However, every permutation can be *built* out of cycles, in the following sense.

**Theorem 16.6** Any permutation  $f \in S_n$  is a composite of disjoint cycles, that is,  $f = f_1 \cdots f_k$  for some  $k \geq 1$  and disjoint cycles  $f_1, \dots, f_k$ .

When  $f$  is written in such a way, we say it is written in *cycle notation*, and call  $f_1 \cdots f_k$  the *cycle decomposition* of  $f$ . Note that the identity  $\iota$  is a product of  $n$  1-cycles  $\iota = (1)(2) \cdots (n)$ .

*Proof.* We prove by strong induction that for  $k = 0, 1, \dots, n$ , if  $f \in S_n$  is a permutation with  $k$  non-fixed points, then  $f$  can be written as a composite of disjoint cycles.

If  $k = 0$  then  $f$  has no non-fixed points, that is, every point is fixed. So  $f = \iota$ , which is a composite of  $n$  disjoint 1-cycles.

Now let  $1 \leq K \leq n$  and suppose we have shown that every permutation in  $S_n$  with fewer than  $K$  non-fixed points is a composite of disjoint cycles. Let  $f \in S_n$  be a permutation with  $K$  non-fixed points.

Since  $K \geq 1$ , we can choose a non-fixed point  $a$ . By the Pigeonhole Principle, the elements  $a, f(a), f^2(a), \dots$  of  $\{1, \dots, n\}$  cannot all be distinct (since  $\{1, \dots, n\}$  is finite), so  $f^i(a) = f^j(a)$  for some integers  $j > i \geq 0$ , giving  $f^{j-i}(a) = a$  with  $j-i \geq 1$ . So, there is a positive integer  $r$  such that  $f^r(a) = a$ .

Let  $r$  be the *smallest* positive integer such that  $f^r(a) = a$ . Write  $a_1 = a$ ,  $a_2 = f(a)$ ,  $\dots$ ,  $a_r = f^{r-1}(a)$ . Then  $f(a_1) = a_2$ ,  $f(a_2) = a_3$ ,  $\dots$ ,  $f(a_{r-1}) = a_r$ , and  $f(a_r) = f^r(a) = a = a_1$ . So

$$f = g \circ (a_1 \ a_2 \ \cdots \ a_r)$$

where  $g \in S_n$  is defined by

$$g(b) = \begin{cases} b & \text{if } b \in \{a_1, \dots, a_r\} \\ f(b) & \text{if } b \notin \{a_1, \dots, a_r\}. \end{cases}$$

Now  $a$  is not a fixed point of  $f$ , so  $r \geq 2$ , which implies that  $g$  has fewer non-fixed points (more fixed points!) than  $f$ . So by inductive hypothesis,  $g$  is a composite of disjoint cycles. Each of these cycles is disjoint from  $(a_1 \ a_2 \ \cdots \ a_r)$ , since  $g(b) = b$  for all  $b \in \{a_1, \dots, a_r\}$ . Thus,  $f$  is a composite of disjoint cycles, completing the induction.

Hence the theorem holds by the principle of mathematical induction. ■

How do we find the cycle decomposition in practice? The proof above gives us a roadmap. Given a permutation  $f$ , we know there is some integer  $r$  such that  $f^r(1) = 1$ . If we write  $a_i = f^i(1)$  then  $(1a_1 \cdots a_{r-1})$  is one of the cycles in the decomposition in  $f$ . To find

another cycle, pick  $k \notin \{1, a_1, \dots, a_{r-1}\}$  and repeat the process: there is some  $s$  such that  $f^s(k) = k$ , and if we write  $b_i = f^i(k)$  then another cycle in the decomposition is  $(kb_1 \cdots b_{s-1})$ . Repeat this process until each integer is in one of the cycles you've found.

**Example 16.9** Let  $f = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 4 & 6 & 1 & 5 & 3 & 2 \end{pmatrix} \in S_6$ . Then

$$f(1) = 4, f(4) = 5, f(5) = 3, f(3) = 1,$$

so one of the cycles in the cycle decomposition of  $f$  is  $(1\ 4\ 5\ 3)$ . Now we look at the cycle that contains 2 (where we picked 2 because it is not one of the integers we have placed in a cycle so far), we see

$$f(2) = 6, f(6) = 2,$$

which implies that  $f = (1\ 4\ 5\ 3)(2\ 6)$ . Recall that disjoint cycles commute, so we could also write  $f$  as  $(2\ 6)(1\ 4\ 5\ 3)$ .

If the cycle decomposition consists of some 1-cycles like  $(j)$ , note that these are just the identity mapping, so we can omit them when writing a permutation in cycle notation. For example, the permutation on  $S_{10}$  that just switches the numbers 1 and 2 can be written succinctly as  $(1\ 2)$  instead of  $(1\ 2)(3)(4)(5)(6)(7)(8)(9)(10)$ .

Recall that *disjoint* cycles commute, but composition of cycles that aren't disjoint are not, in general, commutative.

**Example 16.10** Let  $f = (1\ 2\ 3)(4\ 5)$  and  $g = (1\ 2\ 3\ 4)$  be permutations in  $S_5$ . To find the cycle decomposition of  $fg$ , we do as before: start with 1 and look at the values we get by repeatedly plugging into  $fg$ :

$$fg(1) = f(g(1)) = f(2) = 3,$$

$$fg(3) = f(g(3)) = f(4) = 5,$$

$$fg(5) = f(g(5)) = f(5) = 4,$$

$$fg(4) = f(g(4)) = f(1) = 2,$$

$$fg(2) = f(g(2)) = f(3) = 1.$$

So the first cycle is  $(1\ 3\ 5\ 4\ 2)$ . Note that all numbers in  $\{1, 2, \dots, 5\}$  have been used, and so  $fg = (1\ 3\ 5\ 4\ 2)$ . Now we do the same for  $gf$ , starting with 1 again:

$$gf(1) = g(f(1)) = g(2) = 3,$$

$$gf(3) = g(f(3)) = g(4) = 1,$$

$$gf(2) = g(f(2)) = g(3) = 4,$$

$$gf(4) = g(f(4)) = g(5) = 5,$$

$$gf(5) = g(f(5)) = g(1) = 2$$

and since this uses up all numbers from 1 to 5, we get  $gf = (1\ 3\ 2\ 4\ 5)$ . So  $gf \neq fg$ .

For  $f \in S_n$ , the *cycle shape* of  $f$  is the sequence of numbers we get by listing the orders of the cycle decomposition in decreasing order. If a number appears several times in the sequence, we replace it with that number to the power of how many times it appears.

- Example 16.11** • The permutation  $f = (1\ 4\ 5\ 3)(2\ 6)(7) \in S_7$  has cycle shape  $(4, 2, 1)$ .
- The permutation  $g = (1\ 2)(3\ 4) \in S_4$  has cycle shape  $(2, 2)$ , or more succinctly,  $(2^2)$ .
  - The permutation  $h = (1\ 4)(2\ 3)(5\ 7)(8\ 9\ 11) \in S_{11}$  has cycle shape  $(3, 2^3, 1^2)$ . Why the  $1^2$ ? Recall that 6 and 10 don't appear in any of the cycles in  $h$ , so  $h(6) = 6$  and  $h(10) = 10$ , so really the full cycle decomposition of  $h$  is  $(1\ 4)(2\ 3)(5\ 7)(8\ 9\ 11)(6)(10)$ .

This now lets us calculate the order of any permutation  $f \in S_n$ .

**Lemma 16.7** Let  $f \in S_n$  and let  $m$  be its order. Write

$$f = \sigma_1 \sigma_2 \cdots \sigma_s$$

where  $\sigma_1, \dots, \sigma_s$  are disjoint cycles of lengths  $r_1, \dots, r_s$  respectively. Then

$$m = \text{lcm}(r_1, r_2, \dots, r_s),$$

where  $\text{lcm}$  is the smallest positive integer that is divisible by  $r_i$  for each  $i = 1, \dots, s$ .

*Proof.* Recall that the order of a cycle is just its length, so we know that  $\sigma_i^{r_i} = \iota$ . Let  $d = \text{lcm}(r_1, r_2, \dots, r_s)$  and  $m$  be the order of  $f$ . We wish to show  $m = d$ .

As that disjoint cycles commute, so we can split the following product:

$$f^d = (\sigma_1 \sigma_2 \cdots \sigma_s)^d = \sigma_1^d \sigma_2^d \cdots \sigma_s^d = (\sigma_1^{r_1})^{d/r_1} (\sigma_2^{r_2})^{d/r_2} \cdots (\sigma_s^{r_s})^{d/r_s} = (\iota)^{d/r_1} (\iota)^{d/r_2} \cdots (\iota)^{d/r_s} = \iota.$$

By Lemma 16.5, this means  $m|d$ . However, note that since  $m$  is the order of  $f$ ,

$$\iota = f^m = (\sigma_1 \sigma_2 \cdots \sigma_s)^m = \sigma_1^m \sigma_2^m \cdots \sigma_s^m,$$

again using the fact that the cycles  $\sigma_i$  are disjoint. But the permutations  $\sigma_1^m, \dots, \sigma_s^m$  are also disjoint, and their product is the identity, so  $\sigma_i^m = \iota$  for each  $i$ . By Lemma 16.5 again, this means  $r_i|m$  (since  $r_i$  is the order of  $\sigma_i$ ). By the definition of the least common multiple, this means  $d|m$ . Finally,  $m|d$  and  $d|m$  (and both being positive integers) imply  $d = m$ . ■

Using this lemma, you can now always compute  $m$  by finding the cycle shape.

**Example 16.12** The order of the permutation

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 3 & 4 & 6 & 1 & 5 & 2 & 8 & 7 \end{pmatrix} = (1\ 3\ 6\ 2\ 4)(7\ 8) \in S_8,$$

is  $\text{lcm}(2, 5, 1) = \text{lcm}(2, 5) = 10$ , while the order of

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 5 & 6 & 7 & 8 & 1 & 2 & 3 & 4 \end{pmatrix} = (1\ 5)(2\ 6)(3\ 7)(4\ 8) \in S_8,$$

is  $\text{lcm}(2, 2, 2, 2) = 2$ .

## 16.8 Even and odd permutations

The content of this section will be a little easier to talk about with a motivating example. Recall the *9-puzzle*: this is a puzzle that involves a table with 9 numbers and a missing space in a 3 by 3 grid, like so:



$$\begin{array}{ccc} \square & 2 & 3 \\ 1 & 4 & 5 \\ 7 & 8 & 6 \end{array}$$

where the box represents an empty space. The objective is to put all the numbers in order by sliding numbers into the empty space, creating a new empty space, and so on. So for example, the above puzzle can be solved using the following moves:

$$\begin{array}{ccccc} \square & 2 & 3 & 1 & 2 & 3 & 1 & 2 & 3 & 1 & 2 & 3 & 1 & 2 & 3 \\ 1 & 4 & 5 & \mapsto & \square & 4 & 5 & \mapsto & 4 & \square & 5 & \mapsto & 4 & 5 & \square \\ 7 & 8 & 6 & & 7 & 8 & 6 & & 7 & 8 & 6 & & 7 & 8 & 6 \end{array}$$

This puzzle is a 3 by 3 variant of the *15-puzzle* (on a 4 by 4 grid) that was famous in the late 1800s.

Not every arrangement of numbers can be solved. For example, there is no way of solving the puzzle below

$$\begin{array}{ccc} \square & 3 & 2 \\ 1 & 4 & 5 \\ 7 & 8 & 6 \end{array}$$

How do we know that a given arrangement of numbers and a space can be solved? The math in this section will give us a way of figuring this out.

We start with the following lemma.

**Lemma 16.8** Every permutation in  $S_n$  is a product of cycles of length 2.

This is intuitively clear: if you have a line of people and want to rearrange them into a particular order, you can clearly do it by having them move up and down the line. And every time someone moves one place, they switch places with the person next to them (a 2-cycle).

*Proof.* Let  $f$  be a permutation in  $S_n$ . We can also assume that  $f$  is a cycle of length  $r$  since otherwise, any permutation in  $S_n$  is a product of disjoint cycles, and we can just multiply together the respective products of 2-cycles we get.

So suppose  $f = (a_1 a_2 a_3 \dots a_{r-1} a_r)$ . Then

$$f = (a_1 a_2 a_3 \dots a_{r-1} a_r) = (a_1 a_r)(a_1 a_{r-1})(a_1 a_{r-2}) \cdots (a_1 a_3)(a_1 a_2).$$

One can check this by plugging in various values  $a_i$ . If we plug in  $a_3$ , for example, remember that when plugging a number into a composite  $f_1 \cdots f_n$ , we plug the number into  $f_n$  first, then the resulting value into  $f_{n-1}$ , and so on. So in the above product for  $f$ , we first plug  $a_3$  into  $(a_1 a_2)$ , which just returns  $a_3$ . Then we plug that into  $(a_1 a_3)$ , which gives  $a_1$ . Then we plug that into  $(a_1 a_4)$  and we get  $a_4$ . But now  $a_4$  does not appear in any other 2-cycle, and so they all return  $a_4$ . Hence,  $f(a_3) = a_4$ , and we can show similarly that  $f(a_i) = a_{i+1}$  for all  $i$  except  $r$  where  $f(a_r) = a_1$ . ■

Note that the product of cycles in this lemma is far from being unique, and the cycles of order 2 in the lemma are not necessarily disjoint.

We would like to define a permutation to be *even* if it is a product of even number of cycles of order 2, and *odd* if it is a product of odd number of cycles of order 2. However, it is not clear from this definition whether or not a permutation can be both even and odd at the same time. Hence, we will actually define the properties of being even and odd slightly

differently, and then later deduce that even and odd permutations can be written as an even or odd number of 2-cycles.

Before we formalise this definition, we first introduce some new notation, similar to the summation notation  $\sum$ , we're already familiar with.

**Definition 16.5** Given a sequence of values  $a_i$ , we can take a finite product as

$$\prod_{i=1}^n a_i = a_1 \cdot a_2 \cdot \dots \cdot a_n,$$

and an infinite product as

$$\prod_{i=1}^{\infty} a_i = a_1 \cdot a_2 \cdot \dots$$

Our definition of even and odd permutations in  $S_n$  will rely on the following function of  $n$  variables  $x_1, \dots, x_n$ :

$$\Delta(x_1, \dots, x_n) = \prod_{1 \leq i < j \leq n} (x_i - x_j),$$

that is, the product of all pairs  $(x_i - x_j)$  where  $1 \leq i < j \leq n$ . So for example, in  $S_3$ , we have

$$\Delta(x_1, x_2, x_3) = (x_1 - x_2)(x_2 - x_3)(x_1 - x_3).$$

Given a permutation  $f \in S_n$ , we will define  $f(\Delta)$  to be

$$f(\Delta)(x_1, \dots, x_n) = \prod_{1 \leq i < j \leq n} (x_{f(i)} - x_{f(j)}).$$

**Example 16.13** Let  $n = 3$  and consider the permutation  $f = (12) \in S_3$ , then

$$\begin{aligned} f(\Delta)(x_1, x_2, x_3) &= (x_{f(1)} - x_{f(2)})(x_{f(2)} - x_{f(3)})(x_{f(1)} - x_{f(3)}) \\ &= (x_2 - x_1)(x_1 - x_3)(x_2 - x_3). \end{aligned}$$

Notice that  $f(\Delta)$  is actually just  $\pm\Delta$ , since all the permutation does is change the  $(x_i - x_j)$  to either  $(x_i - x_j)$  or  $(x_j - x_i)$ . We define the *sign* of  $f$  to be

$$\text{sgn}(f) = \frac{f(\Delta)}{\Delta},$$

that is,  $\text{sgn}(f) = 1$  if  $f(\Delta) = \Delta$  and  $\text{sgn}(f) = -1$  if  $f(\Delta) = -\Delta$ .

**Definition 16.6** We say a permutation  $f \in S_n$  is *even* if  $\text{sgn}(f) = 1$  and *odd* if  $\text{sgn}(f) = -1$ .

The above definition of even and odd and the sign may seem really technical to work with, but the lemma below will provide us with some short-cuts so that we never actually have to refer to  $n$ -variable polynomials:

**Lemma 16.9** Let  $n \in \mathbb{N}$ .

- (a) The sign of  $\iota$  is 1 and any 2-cycle is  $-1$ .
- (b) For any  $f$  and  $g$  in  $S_n$ , we have  $\text{sgn}(fg) = \text{sgn}(f)\text{sgn}(g)$ .
- (c) The sign of a cycle of length  $r$  is  $(-1)^{r-1}$ .
- (d) For every permutation  $f \in S_n$ , let  $f = \sigma_1 \sigma_2 \dots \sigma_s$  where  $\sigma_1, \dots, \sigma_s$  are disjoint cycles

of lengths  $r_1, \dots, r_s$ . Then

$$\operatorname{sgn}(f) = (-1)^{r_1-1}(-1)^{r_2-1}(-1)^{r_3-1} \dots (-1)^{r_s-1}.$$

(e) If  $f \in S_n$ , then  $\operatorname{sgn}(f) = \operatorname{sgn}(f^{-1})$ .

*Proof.* (a) First,  $\operatorname{sgn}(\iota) = \frac{\Delta}{\Delta} = 1$ . Next, let  $(a \ b)$  be a 2-cycle with  $a < b$ . Then the only terms  $(x_i - x_j)$  in the polynomial  $\Delta$  that change sign (i.e. are reversed) when we compute  $(a \ b)(\Delta)$  are the terms

$$\begin{aligned} (x_a - x_{a+1})(x_a - x_{a+2}) \cdots (x_a - x_b) \\ (x_{a+1} - x_b)(x_{a+2} - x_b) \cdots (x_{b-1} - x_b) \end{aligned}$$

There are  $b - a$  terms on the first line and  $b - a - 1$  on the second line, for a total of  $2(b - a) - 1$ , an odd number, hence

$$(a \ b)(\Delta) = (-1)^{2(b-a)-1} \Delta = -\Delta,$$

so  $\operatorname{sgn}((a \ b)) = -1$ .

(b) Let  $f$  and  $g$  be two permutations. Notice that  $f(-\Delta) = -f(\Delta)$ , and so

$$fg(\Delta) = f(g(\Delta)) = f(\operatorname{sgn}(g)\Delta) = \operatorname{sgn}(g)f(\Delta) = \operatorname{sgn}(g)\operatorname{sgn}(f)\Delta$$

and this implies  $\operatorname{sgn}(fg) = \operatorname{sgn}(f)\operatorname{sgn}(g)$

(c) Now things get a lot easier: If  $f$  is an  $r$ -cycle, then by the proof of Lemma 16.8, it is a product of  $r - 1$  many 2-cycles  $\sigma_1, \dots, \sigma_r$ , so by (a) and (b),

$$\operatorname{sgn}(f) = \operatorname{sgn}(\sigma_1) \cdots \operatorname{sgn}(\sigma_r) = (-1)^{r-1}.$$

(d) The same proof as in (c) proves this one.

(e) Finally,

$$1 = \operatorname{sgn}(\iota) = \operatorname{sgn}(ff^{-1}) = \operatorname{sgn}(f)\operatorname{sgn}(f^{-1})$$

and since the sign is always  $\pm 1$ , we can divide both sides by  $\operatorname{sgn}(f^{-1})$  to get

$$\operatorname{sgn}(f) = \operatorname{sgn}(f^{-1})^{-1} = \operatorname{sgn}(f^{-1}).$$

■

As a corollary of the above lemma, we get the following.

**Corollary 16.10** Let  $f$  be a permutation in  $S_n$ .

1.  $f$  is **even** if and only if it is a product of even number of cycles of order 2.
2.  $f$  is **odd** if and only if it is a product of odd number of cycles of order 2.

**Example 16.14** The permutation

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 1 & 5 & 4 \end{pmatrix} = (1 \ 2 \ 3)(4 \ 5) = (1 \ 3)(1 \ 2)(4 \ 5)$$

is odd, because it is a product of 3 cycles of length 2.

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 4 & 5 & 1 \end{pmatrix} = (1 \ 2 \ 3 \ 4 \ 5) = (1 \ 5)(1 \ 4)(1 \ 3)(1 \ 2) \text{ is even.}$$

**Example 16.15** Let's revisit the 9-puzzle and start with the one we solved earlier, that is, where we wanted to rearrange

$$\begin{array}{ccc} \square & 2 & 3 \\ 1 & 4 & 5 \\ 7 & 8 & 6 \end{array} \text{ into } \begin{array}{ccc} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & \square \end{array}.$$

If we consider the  $\square$  as being 9, the above arrangement on the left corresponds to applying the permutation

$$f = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 4 & 2 & 3 & 5 & 6 & 9 & 7 & 8 & 1 \end{pmatrix} = (14569)(2)(3)(7)(8) = (14569) = (19)(16)(15)(14)$$

to the grid on the right. Note we can write this as the product of 4 2-cycles meaning it is an even permutation.

Performing a move by moving a number into the  $\square$  corresponds to applying a 2-cycle. Let's revisit the moves we did earlier to solve this puzzle and indicate what 2-cycles we are using:

$$\begin{array}{ccc} \square & 2 & 3 \\ 1 & 4 & 5 \\ 7 & 8 & 6 \end{array} \xrightarrow{(14)} \begin{array}{ccc} 1 & 2 & 3 \\ \square & 4 & 5 \\ 7 & 8 & 6 \end{array} \xrightarrow{(45)} \begin{array}{ccc} 1 & 2 & 3 \\ 4 & \square & 5 \\ 7 & 8 & 6 \end{array} \xrightarrow{(56)} \begin{array}{ccc} 1 & 2 & 3 \\ 4 & 5 & \square \\ 7 & 8 & 6 \end{array} \xrightarrow{(69)} \begin{array}{ccc} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & \square \end{array}$$

Thus, applying the permutation  $(69)(56)(45)(14)$  returns the board to the original order, that is, it inverts the permutation  $(14569)$ .

Now let's look at the following puzzle:

$$\begin{array}{ccc} \square & 3 & 2 \\ 1 & 4 & 5 \\ 7 & 8 & 6 \end{array} \text{ into } \begin{array}{ccc} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & \square \end{array}.$$

The permutation corresponding to this arrangement is

$$g = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 4 & 3 & 2 & 5 & 6 & 9 & 7 & 8 & 1 \end{pmatrix} = (14569)(23)(7)(8) = (14569)(23)$$

Notice that if the puzzle can be solved, that requires moving the square up or down an even number of times, and left and right an even number of times. Thus, the puzzle must take an even number of moves, and since each move is a 2-cycle, that implies that the inverse of the above permutation must be even, and hence the permutation itself must also be even (this uses Lemma 7.10 (a),(d) and (e)). However, the sign of the above permutation (using Lemma 7.10 (b) and (c)) is

$$\text{sgn}((14569)(23)) = \text{sgn}((14569))\text{sgn}((23)) = (-1)^{5-1}(-1)^{2-1} = -1.$$

and so this is an odd permutation, and this gives a contradiction. Thus, the original puzzle is not solvable.

## 16.9 Exercises

The relevant exercises to this section are in Chapter 20 of Liebeck.

■ **Exercise 16.3** Consider the permutation

$$f = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 3 & 5 & 4 & 8 & 7 & 6 & 2 & 1 \end{pmatrix}$$

Write  $f$ ,  $f^2$ ,  $f^{-1}$ , and  $f^{146}$  in cycle notation.

■ **Exercise 16.4** Given  $2 \leq k \leq n$ , how many  $k$ -cycles are there in  $S_n$ ?

■ **Exercise 16.5** How many order 3 permutations are there in  $S_4$ ?

■ **Exercise 16.6** How many order 3 permutations are there in  $S_6$ ?

■ **Exercise 16.7** We say a set  $S \subseteq S_n$  *generates*  $f \in S_n$  there are  $f_1, \dots, f_k \in S$  so that  $f = f_1 \cdots f_k$  (where we could have  $k = 1$ , that is, this could just be a composition of just one function), and  $S$  *generates*  $S_n$  if it generates  $f$  for every  $f \in S_n$ .

1. Show that if  $f \in S$ , then  $S$  generates  $\iota$  and  $f^{-1}$ .
2. Show that  $S = \{(1\ 2), (2\ 3), \dots, (n-1\ n)\}$  generates  $S_n$ .
3. Show that  $T = \{(1\ 2), (1\ 2\ 3), \dots, (1\ 2 \dots n)\}$  generates  $S_n$ .

■ **Exercise 16.8** Given  $x, y \in S_n$ , we call  $z = xyx^{-1}$  the *conjugate of  $y$  by  $x$* . If  $z = xyx^{-1}$  for some  $x \in S_n$ , we say  $z$  is a *conjugate* of  $y$ .

- (a) Show that the relation  $y \sim z$  if  $y$  is conjugate to  $z$  is an equivalence relation on  $S_n$ .
- (b) Fix  $x \in S_n$ . Show that the function sending  $y \in S_n$  to  $xyx^{-1}$  is a bijection on  $S_n$ .
- (c) If  $c = (a_1\ a_2 \dots a_k)$  is a cycle in  $S_n$  and  $x \in S_n$ , prove that

$$xcx^{-1} = (x(a_1) \dots x(a_k)).$$

(d) If  $y, z$  are conjugates, what can you say about their cycle shapes? Is the converse also true?

■ **Exercise 16.9 Landau's function**  $L(n)$  is defined for every  $n \in \mathbb{N}$  to be the largest order of an element of  $S_n$ . This has no convenient formula, but we will prove some simple bounds on it in this exercise.

- (a) Show that for all  $n \in \mathbb{N}$ ,  $L(n) < 2^n$ . *Hint: it may help to show  $n < 2^n$  for all  $n \in \mathbb{N}$ .*
- (b) Show that  $L(n) \geq \frac{n}{2} \cdot \left(\frac{n}{2} - 1\right)$  for every  $n \in \mathbb{N}$ . *Hint: Remember that  $L(n)$  is the maximum order of an element in  $S_n$ , so to prove the above claim, you just need to find one element of  $S_n$  with order at least  $\frac{n}{2} \cdot \left(\frac{n}{2} - 1\right)$ . Try playing around with different products of cycles to see how big you can make their order.*